

# Abhinav Kesarwani

Generative AI | LLM Engineering | Machine Learning | Software Engineering | Data Science

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## EDUCATION

### Gati Shakti Vishwavidyalaya

B.Tech. in Artificial Intelligence & Data Science

- **CGPA:** 8.89/10

Vadodara, Gujarat

Aug 2023 – Aug 2027

## TECHNICAL SKILLS

**Programming Languages:** Python, C++, SQL

**Generative AI:** RAG, LLM Fine-tuning (LoRA), Prompt Engineering, Transformer Models, Semantic Search

**Frameworks:** PyTorch, Hugging Face Transformers, LangChain, FastAPI, Flask, Scikit-learn

**LLM & Retrieval:** FAISS, BM25, Sentence Transformers, Cross-Encoder Reranking, OCR

**Deployment:** AWS, Docker, Kubernetes, GitHub Actions, CI/CD

**Database & Tools:** MySQL, Git

## PROFESSIONAL EXPERIENCE

### Siemens

Technical Intern

Pune, Maharashtra

May 2026 – August 2026

- Developed Python-based automation solutions that streamlined engineering workflows, reducing manual effort by **6–7 hours per delivery**, and currently engineering an AI-powered intelligent code review platform to enhance software quality in collaboration with cross-functional engineering teams.

### Delhi Metro Rail Corporation (DMRC)

IT Intern

New Delhi

May 2025 – Jun 2025

- Developed and deployed an AI-powered Indoor Navigation System for the DMRC Academy, currently supporting navigation for **100+ daily visitors**, while automating trip-chart generation and gaining hands-on experience with enterprise-scale transportation systems.

## PROJECTS

### SecureDocAI — Privacy-Preserving Offline RAG Framework

Jan 2025 – Present

- Built a fully offline enterprise-grade **RAG** framework using FAISS, BM25, Cross-Encoder reranking, OCR, Sentence Transformers, and local SLMs to perform semantic question answering over **35K+ indexed document chunks** from **25+ heterogeneous documents**.
- Implemented a modular retrieval pipeline with hybrid dense-lexical search, OCR processing, transformer-based reranking, and interchangeable LLM components; benchmarked the system across **7 ablation configurations** using retrieval, answer quality, latency, and faithfulness metrics.
- Designed a comprehensive evaluation suite containing **125 manually verified benchmark questions** and **15+ evaluation metrics**; proposed the **Evidence-Based Confidence Score (EBCS)** to improve transparency and reliability of offline AI-generated responses.

### AI Code Reviewer — Production-Ready AI Code Review Platform

 GitHub

- Engineered a cloud-native AI code review platform leveraging **GPT-4o**, GitHub Apps, and intelligent analysis pipelines to automatically identify security vulnerabilities, architectural anti-patterns, code quality issues, and best-practice violations in pull requests.
- Architected a cloud-native platform comprising **5 microservices** using **AWS EKS**, Docker, Kubernetes, PostgreSQL, Redis, asynchronous workers, and GitHub Actions CI/CD for automated enterprise-scale code analysis.
- Integrated observability with LangFuse, Prometheus, and Grafana while implementing modular AI inference, repository management, review orchestration, and deployment pipelines to improve maintainability and production monitoring.

## RESEARCH

- **Accident Analysis and Safety Improvements in Indian Railways** accepted at ICICST 2026, to appear in SSRN Proceedings (Elsevier).

## ACHIEVEMENTS

- Secured **2nd Place** in **two university-level hackathons** for AI-powered transportation and enterprise software solutions.
- Awarded **3rd Rank** in the **HCL GUVI Scholarship Program**.
- Conducted **Power BI workshops** for **40+ Indian Air Force officers** on data visualization and decision-support analytics.

## LEADERSHIP & CERTIFICATIONS

- **Leadership:** Placement Coordinator, Internship Coordinator, Marketing & PR Co-Coordinator Business Club
- **Certifications:** NVIDIA — Applications of AI for Predictive Maintenance, Fundamentals of Deep Learning, Computer Vision for Industrial Inspection; IBM — Artificial Intelligence Fundamentals.